United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam: Spring 2025
Course Code: CSE 3811, Course Title: Artificial Intelligence
Total Marks: 40 Duration: 2 hours

Answer all questions. Marks are indicated on the right side of each question.

[**Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.**]

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1. Four friends, Ryomen (R), Satoru (S), Yuji (Y), and Mahito (M) have come to a restaurant near their university. The restaurant serves Chicken Burger (C), Beef Burger (B), Kashmiri Naan (K), and Paratha (P). Their preferences are as follows:
i. Ryomen does not like Paratha.
ii.. Satoru and Yuji would like to grab a bite of each other's food, so they will order different dishes.
iii. Yuji likes burgers, so he will order a Chicken Burger or a Beef Burger.
iv. Ryomen likes to copy Satoru, so he will order the same dish as Satoru.
v. Mahito will not order Kashmiri Naan because he had them earlier.
vi. Mahito is obsessed with being unique, so he will not order the same dish as anyone else.
Based on these constraints, answer the following questions:
a. Draw the constraint graph for this CSP. **[2]**
b. Solve the CSP using **MRV and LCV**. In case of any ties, choose the **alphabetically smaller** option. **[6]**

2. A survey was conducted on **300** Pokémon fans to analyze their preferences regarding Pokémon types. Half of the participants were **adults**, and the other half were **young** fans. Among the adults, the ratio of males to females is **2:1**, whereas among the young fans, **50%** are males and the rest are females.
Among male adults, the preference distribution among **Fire-type**, **Water-type**, and **Grass-type** Pokémon follows the ratio **4:4:2**. Among female adults, the preference ratio is **3:3:4**. Among young male fans, the type preference is **2:2:1**, and among young female fans, the preferences for Fire, Water, and Grass types are equally distributed.

a. Based on this data, construct the **full joint distribution table** over the three random variables: Category (C) ∈ {Adult, Young}, Gender (G) ∈ {Male, Female} and Type (T) ∈ {Fire, Water, Grass} **[3]**
b. Find the **probability** that a randomly selected fan is a young male who prefers Fire-type Pokémon. **[2]**
c. If a fan is randomly picked, and found to be a female young fan, what is the **probability** that she loves Water-types. **[3]**

3. (a) In a peer-reviewed academic conference, the Paper Acceptance decision is influenced by two main factors: **Reviewer Scores** and **Reputation of the Authors**.

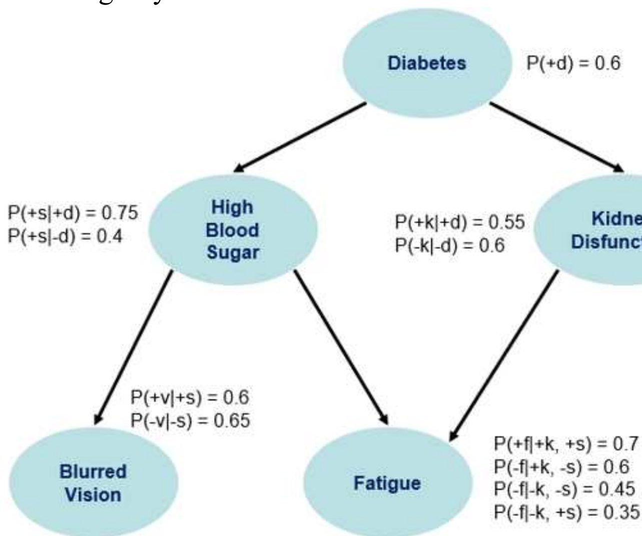
- A **Well-Written Paper** increases the likelihood of high Reviewer Scores.
- Reviewer Scores also depend on the **Novelty of Findings**.
- A **Strong Publication History** increases the Author Reputation.

If a paper is accepted:

- It is more likely to be **Cited Early** after publication.
- The Author is **invited as a Speaker** at the next event.

I. Construct the Bayes' Net from the above description. **[2]**
II. Compare the memory complexity of storing the entries for Full Joint Distribution and the CPT of Bayes' Net. **[2]**

(b) Consider the following Bayes' Net:



Find the probability of a person having fatigue given he has diabetes using Inference by Enumeration. **[4]**

4. Dr. Eliza Thornberry is researching animal behaviors in various environments. The following dataset logs observed features and the outcome behavior of animals in each observation. Use a Naive Bayes Classifier with **Laplace smoothing (k = 2)** to determine the most likely behavior outcome for a random animal event with attributes: **<Evening, Mountain, Heavy Rain, Noisy>** **[8]**

Obs No	**Time**	**Terrain**	**Weather**	**Sound**	**Behavior**
1	Morning	Forest	Sunny	Quiet	Playful
2	Afternoon	Mountain	Sunny	Quiet	Playful
3	Morning	Desert	Windy	Noisy	Sleepy
4	Evening	Forest	Heavy Rain	Noisy	Playful
5	Night	Desert	Windy	Quiet	Calm
6	Evening	Mountain	Snowy	Noisy	Calm
7	Afternoon	Forest	Windy	Quiet	Sleepy
8	Evening	Mountain	Heavy Rain	Quiet	Playful
9	Morning	Mountain	Snowy	Quiet	Calm
10	Night	Forest	Heavy Rain	Noisy	Sleepy
11	Afternoon	Desert	Sunny	Quiet	Playful
12	Night	Forest	Snowy	Quiet	Calm
13	Morning	Desert	Sunny	Noisy	Playful
14	Afternoon	Mountain	Windy	Quiet	Playful
15	Evening	Forest	Sunny	Noisy	Calm

5. An insurance company aims to predict whether an applicant is at risk of developing heart disease based on their age group, smoking habits, and exercise routine before approving health policies. Suppose you have been hired as a data scientist to develop a Decision Tree model for this task. The dataset provided is as follows:

Serial	**Age Group**	**Smoking**	**Exercise**	**Heart Disease**
0	**middle**	**yes**	**regular**	**yes**
1	**middle**	**yes**	**occasional**	**yes**
2	**old**	**yes**	**occasional**	**yes**
3	**old**	**yes**	**none**	**yes**
4	**old**	**yes**	**none**	**yes**
5	**old**	**yes**	**regular**	**no**
6	**young**	**no**	**none**	**yes**
7	**young**	**no**	**occasional**	**no**
8	**young**	**no**	**occasional**	**no**
9	**young**	**no**	**none**	**no**
10	**middle**	**no**	**regular**	**no**
11	**middle**	**no**	**regular**	**No**

a. Determine the root node of the Decision Tree by calculating the Information Gain for each attribute (Age Group, Smoking, Exercise). Show your calculations. **[6]**

b. After selecting the root node using the highest Information Gain, the data is split based on this attribute, creating a simple Decision Tree with one decision level. For each resulting leaf node, the probability of a "Heart Disease = yes" outcome can be calculated using the formula:

$$P(Heart\ Disease\ =\ yes)\ =\ \frac{count(Heart\ Disease = yes)}{count(Heart\ Disease = yes)+count(Heart\ Disease = no)}$$

Draw the resulting simple Decision Tree (with depth=1) based on your chosen root attribute, and calculate P(Heart Disease = yes) for the individual with following characteristics:

Age Group = young, Smoking = yes, Exercise = regular **[2]**